

ORACLE ENGINEER HANDBOOK

The **ROWNUM** Paradox

Why "WHERE ROWNUM > 1" always returns zero rows and how to master the evaluation lifecycle.

PART 01

The Mechanism

Understanding how the Oracle Optimizer processes pseudocolumns during row fetch.

The Row Evaluation Lifecycle



1. Fetch

Oracle fetches the first candidate row from the source table or index scan.



2. Assign

The engine assigns `ROWNUM = 1` to this specific row before any filtering.



3. Evaluate

The `WHERE` clause is tested. If it fails, the row is discarded immediately.



4. Increment

The counter only increments to 2 if the current row successfully passed the filter.

| The "Greater Than One" Trap

The Problem Query

```
SELECT * FROM employees WHERE ROWNUM > 1;
```

Logic: The first row gets ROWNUM 1. Since 1 is not > 1, the test fails. The row is deleted and ROWNUM 1 is reused for the next candidate.

The Result

0 ROWS SELECTED

The counter never reaches 2 because row 1 never survives. This cycle repeats infinitely across every row in the table.

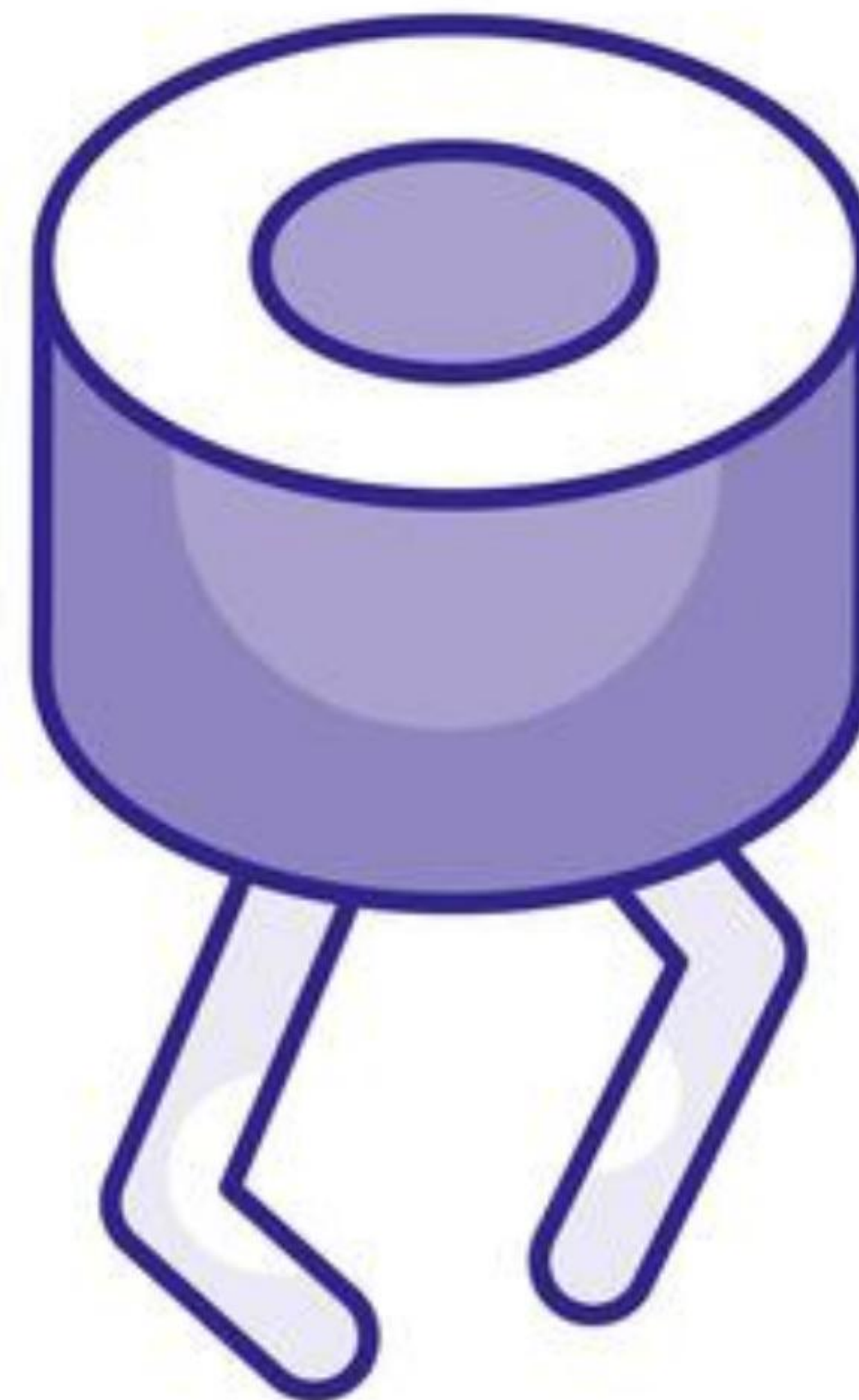
Visualizing the Loop

The Conveyor Belt: Think of data rows as items on a belt.

The Stamper: Every new item gets stamped with "1" by default.

The Trapdoor: Your filter acts as a trapdoor. Because "1" fails `" > 1"`, the item drops.

Reset: The next item arrives, gets the same "1" stamp, and falls too.



PART 02

The Solutions

Strategies for effective pagination and range-based filtering.

Correct Patterns for Row Limits



Inline Views

Wrap the query in a subquery to "materialize" the ROWNUM into a static column value.



OFFSET/FETCH

Use modern SQL syntax (12c+) to handle pagination without manual counter management.



ROW_NUMBER()

Use Analytic functions for complex window-based partitioning and sorting requirements.

Syntax Evolution Comparison

Feature	Classic ROWNUM	ROW_NUMBER()	OFFSET / FETCH
Complexity	High (Nested Views)	Medium	Low (Standard)
Pagination Range	Requires Alias	Native Support	Native Support
Performance	Excellent (Pushdown)	Variable	Excellent (Internal)
Oracle Version	All Versions	8i+	12c+

Performance: Pagination Methods



Benchmarks based on 1M rows table without index support. ROWNUM based pagination benefits from early-exit optimization.

| Best Practices for Pagination

- ✓ **Always Sort First:** ROWNUM is assigned before `ORDER BY`. Use an inline view to sort, then apply limits.
 - ⚙️ **Index Match:** Ensure the `ORDER BY` columns are indexed to allow the optimizer to use "Top-N" pushdown.
 - ⊘ **Avoid Direct Filters:** Never use `>` or `BETWEEN` directly on ROWNUM pseudocolumns.
 - 🔗 **Modern Preference:** Use `FETCH FIRST n ROWS ONLY` for readable, maintainable code.
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"In Oracle, ROWNUM is a phantom. It haunts every fetch loop, becoming real only when captured inside a subquery."

— Oracle Engineering Handbook

Q&A

Deep Dives into ROWNUM Internals

Database Internals Group

oracle-internal-labs.example.com

| Image Sources



<https://static.vecteezy.com/system/resources/thumbnails/071/187/042/large/get-this-amazing-isometric-icon-of-robotic-arm-premium-video.jpg>

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